

Part 2 of 2

Flexible Electronics: What's the Use

Having dealt with the materials used for flexible electronics, types of flexible circuits and their integration process in first part of this article, let us now focus on the applications of flexible circuits

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Why do we need another form of electronics? Standard silicon-chip-based electronics is making great strides in miniaturisation, cramming more functionality into smaller chips. Couldn't we just embed these chips into flexible materials to create flexible applications?

Answer is No. New applications like organic light-emitting diodes (OLEDs) and organic photovoltaics (OPVs) need big areas. "With silicon-based circuits, you would need huge numbers of embedded chips to get the necessary intensity and uniformity. That would be prohibitively expensive and would generate so much heat locally, you would melt the substrate!"

Flexible electronics could transform the way we make and use electronic devices. Flexible electronics opens the door to foldaway smartphone displays, solar cells on a roll of plastic and advanced medical devices—if we can figure out how to make them. Nearly everyone knows what the inside of a computer or a mobile phone looks like: A stiff circuit board, usually green, crammed with chips, resistors, capacitors and sockets, interconnected by a suburban sprawl of printed wiring. What if our printed circuit board was not stiff and was flexible enough to bend or even fold?

Why flexible electronics

Flexible electronics is in vogue for two reasons. First, it promises an entirely new design tool. Imagine, for example, tiny smartphones that wrap around our wrists, and flexible displays that fold out as large as a television. Or photovoltaic cells and reconfigurable

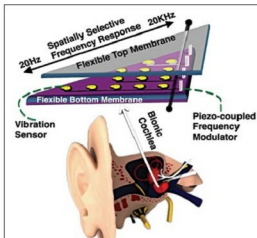


Fig. 10: A bionic ear

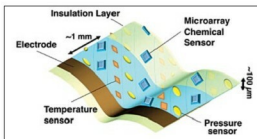


Fig. 11: Schematic diagram of the active-sensor-matrix thin film for the artificial nose and tongue

antennae that conform to the roofs and trunks of our cars. Or flexible implants that can monitor and treat cancer or help paraplegics walk again.

Second, flexible electronics might cost less to make. Conventional semiconductors require complex processes and multi-billion dollar foundries. Researchers hope to print flexible electronics on plastic film the same way we print ink on newspapers. "If we could make flexible electronics cheap enough, you could have throwaway electronics. You could wear your phone on your clothing, or run a bioassay to assess your health simply by wiping your nose with a tissue," Gomez says.

Before any of this happens, though, researchers have to rethink what they know about electronics. None of the flexible electronic devices under development would match the billions of transistors that now fit on silicon chips, or their billions of on-off cycles per second. They would not have to. Despite the challenge, flexible electronics promises changes that go beyond folding displays, inexpensive solar cells, antennae and sensors. It could veer off in some unexpected directions, such as helping paraplegics walk again.

Applications in health care

Flexible electronics has attracted a lot of attention for its enormous potential in many important applications, such as wearable health monitoring devices and medical implants.

While a number of approaches to making flexible sensors or electronics have been developed over the last two decades, flexibility in electronic materials is very attractive for medical and bioengineering. Living organisms are intrinsically flexible and malleable. Thus, flexibility is a necessity for successful integration of electronics in biological systems. Furthermore, in order to carry out daily tasks, flexibility is less likely to hinder over stiffness.

Recently, some electronic devices have been integrated into human bodies. One example is the bionic eye. Here a vision-compromised patient requires an electrically-active addressable matrix array, with each unit or